Contrastive Knowledge Distillation of Unfolded ADMM Self-Representation for Scalable Hyperspectral Image Clustering

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Abstract

Deep subspace clustering has achieved strong performance for hyperspectral image (HSI) clustering, yet most methods rely on explicitly optimizing an $N \times N$ self-representation matrix, leading to quadratic computation/memory and many free variables that are prone to overfitting. To this end, we propose a scalable HSI clustering framework via Contrastive Knowledge Distillation from an unfolded ADMM self-representation solver. We construct a teacher–student paradigm where the teacher is an unfolded ADMM network that computes batch-wise self-representation coefficients efficiently with a few learnable parameters. Based on the teacher coefficients, we select pairs with large representation weights as positives, and pairs with small weights and low feature cosine similarity as negatives, which serve as supervisory signals to train a lightweight student classification network using a contrastive objective. The model is trained in two stages: we first train the unfolded teacher, then freeze it to guide contrastive distillation of the student under a mini-batch strategy, enabling scalability to large datasets. Experiments on three benchmark HSI datasets demonstrate that the proposed method consistently outperforms state-of-the-art clustering approaches.

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Index Terms

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) provide rich spectral signatures across hundreds of contiguous bands, enabling fine-grained material discrimination beyond traditional multispectral imagery [1], [2]. Such a high spectral resolution has supported a wide range of applications, including precision agriculture [3], environmental monitoring [4], military defense [5], and urban planning. In these scenarios, extracting accurate and efficient semantic information from HSIs remains both critical and challenging.

Existing HSI analysis pipelines typically rely on supervised classification [1] or semi-supervised learning [6], [7], which can achieve remarkable performance but require extensive and costly annotations, limiting their practicality [8]. This has motivated increasing interest in unsupervised clustering-based solutions [9]–[13], which exploit intrinsic spectral similarity without labeled data.

Existing clustering techniques generally fall into conventional and deep learning-based approaches [14]. Conventional methods, including centroid/prototype-based [15]–[18], density-based [19]–[21], graph- and spectral-based [22]–[26], hierarchical [27], [28], and subspace clustering [8], [29]–[34], typically rely on handcrafted affinity construction and optimization defined on the original input data (or their induced graphs/representations). While often interpretable and computationally lightweight, fixed similarity modeling and shallow representations make it difficult to cope with nonlinear mixing effects, spatial heterogeneity, and noise in real-world HSIs. Among them, subspace approaches can partially alleviate these issues by exploiting the union-of-linear-subspaces prior, which often yields more robust clustering in high-dimensional spectral spaces. However, their intrinsic linear subspace assumption and the separation between affinity construction and

representation learning still limit their ability to model highly nonlinear and complex spectral–spatial interactions.

To address these limitations, deep learning-based clustering methods have gained increasing attention due to their stronger ability to learn nonlinear and coupled spectral–spatial representations. Broadly, existing deep clustering approaches can be categorized into purely data-driven frameworks [35]–[42], which learn data-adaptive representations with minimal explicit priors and perform clustering either in a two-stage manner (representation learning followed by conventional clustering) or via joint optimization, and model-aware deep clustering methods [43]–[50], which integrate classical clustering principles into neural architectures to improve structure preservation and robustness.

Within the model-aware family, deep subspace clustering has become a representative direction for HSI clustering, coupling deep representation learning with subspace clustering priors to better model spectral–spatial patterns while preserving subspace-consistent structures. However, most self-representation-based deep subspace clustering methods still rely on learning dense inter-sample affinities, incurring $\mathcal{O}(N^2)$ time/memory complexity [43], [47], [50], where N is the dataset size. Moreover, dense affinity learning typically yields a highly over-parameterized formulation (with parameters/variables scaling with N^2), which can easily overfit to dataset-specific correlations and degrade robustness under noise and complex real-world conditions. To improve scalability, basis-style alternatives reduce the computational burden by replacing point-to-point affinities with basis-mediated representations, but may weaken direct inter-point relationships and compromise clustering quality on complex data [44], [51].

To tackle these challenges, we propose a scalable HSI clustering framework based on a teacher–student design that couples an unfolded ADMM self-representation teacher with contrastive learning in a lightweight student network. Instead of constructing a global $N \times N$ self-representation matrix, the teacher performs within-batch self-representation and outputs an $M \times M$ coefficient matrix per mini-batch, reducing the memory footprint

from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$, where M is the batch size. The dominant cost per mini-batch typically comes from solving an $M \times M$ linear system (e.g., inversion/factorization), leading to $\mathcal{O}(M^3)$ time in the worst case; since $M \ll N$ and is fixed in practice, the overall per-epoch complexity scales approximately linearly with N.

To preserve global clustering information beyond batch boundaries, we introduce a contrastive knowledge distillation strategy. We first train the unfolded ADMM network and then use it as an inference-only teacher to mine reliable positive/negative pairs from batch-level self-representations; these relational cues are distilled into a lightweight student classification network via a contrastive objective. By repeatedly sampling mini-batches across epochs, the student progressively consolidates local structural relations into a coherent global embedding, enabling efficient large-scale clustering; after training, only the student is used for fast inference and clustering.

The main contributions of this paper are summarized as follows:

- We introduce a scalable mini-batch-based deep subspace clustering framework that avoids the quadratic dependence on the full dataset size N inherent in global self-representation, making it suitable for large-scale HSIs.
- We design an unfolded ADMM network integrated with autoencoder-based feature extraction, facilitating end-to-end optimization of within-batch self-representation and latent representations.
- We develop a contrastive knowledge distillation framework to transfer batch-level structural relations into a global student network, strengthening the modeling of global spectral–spatial relational cues for clustering.
- Extensive experiments on three benchmark datasets demonstrate our method's superior clustering accuracy and computational efficiency compared to state-of-the-art approaches.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the proposed methodology. Experimental results and analyses are

presented in Section IV. Finally, Section V concludes the paper.

II. Related Work

In this section, we review the existing literature relevant to our research, focusing on subspace clustering, unfolding-based optimization methods, and contrastive clustering.

A. Subspace Clustering

Subspace clustering has attracted significant attention in hyperspectral image clustering due to its capability to identify low-dimensional subspaces underlying high-dimensional spectral data. Two widely studied methods are Sparse Subspace Clustering (SSC) [29] and Low-Rank Representation (LRR) [52], both built upon the self-representation paradigm. In these models, each data point is represented as a linear combination of other samples in the dataset, and the resulting coefficients are expected to be subspace-preserving, yielding an $N \times N$ self-representation matrix.

To enhance clustering performance, various extensions have been proposed. Huang *et al.* [8] and Zhang *et al.* [30] incorporate spatial–spectral structures to improve clustering accuracy, while Huang *et al.* [53] and Traganitis *et al.* [54] employ sketching techniques to improve computational efficiency. Additionally, dictionary learning-based approaches [31], [32] have been explored to enhance scalability. Despite these advancements, standard subspace clustering formulations rely on *linear* self-representation and can be inadequate for data exhibiting complex nonlinear structures.

To address complex and nonlinear data distributions, deep subspace clustering methods have emerged by embedding the self-representation framework into deep neural networks, typically autoencoders. These approaches integrate nonlinear representation learning with subspace priors, making linear self-representation more effective in the learned latent space. Ji *et al.* [43] proposed Deep Subspace Clustering Networks (DSC-Net), which jointly optimize latent feature extraction and self-representation learning. Subsequent

studies further incorporate enhanced learning strategies, such as self-supervised learning by Li *et al.* [47], to refine clustering quality. Moreover, stronger feature extractors and network designs have been explored by Valanarasu *et al.* [50] and Chen *et al.* [55], improving representation capacity. Related model-aware deep clustering models also investigate structured relational modeling (e.g., hypergraph-structured regularization) to exploit high-order dependencies [45].

However, *self-representation* deep subspace clustering models typically require learning a dense $N \times N$ self-representation matrix, which severely limits scalability to large-scale datasets. To alleviate this limitation, scalable alternatives avoid constructing the full pairwise matrix by learning compact basis representations, improving efficiency in large-scale scenarios [44], [51]. Nevertheless, such designs may weaken direct inter-point relationships, potentially compromising clustering performance when the data exhibit complex structures.

B. Unfolding-Based Optimization Methods

Unfolding-based optimization methods have shown strong performance in a variety of imaging inverse problems, including image denoising [56]–[58], image super-resolution [59], [60], and compressive sensing [61], [62]. These approaches transform traditional iterative optimization algorithms into neural networks by mapping each update step to a network layer, where learnable parameters replace hand-crafted update rules. This principled design preserves the interpretability of the original optimization while enabling data-driven adaptation, often leading to improved reconstruction quality and robustness. More recently, unfolding techniques have also been introduced into subspace clustering, achieving promising results [63]. Nevertheless, extending unfolding-based clustering to large-scale datasets remains challenging, since increasing the number of unfolding steps raises computational cost, and dense self-representation matrices can incur prohibitive time/memory overhead.

C. Contrastive Clustering

Contrastive clustering has recently gained traction as an effective paradigm for learning discriminative representations by contrasting positive and negative pairs. In essence, it pulls semantically similar samples closer in the embedding space while pushing dissimilar ones apart. This idea is closely related to self-supervised contrastive representation learning, exemplified by Simple Contrastive Learning of Representations (SimCLR) [64], where positive pairs are often constructed from different augmented views of the same instance. More generally, such views can be obtained either through data augmentation or from naturally available multi-view/multi-modal observations.

Building on this principle, several contrastive clustering methods have been developed. Li *et al.* [65] explicitly perform both instance-level and cluster-level contrastive learning by jointly optimizing the corresponding objectives. Zhong *et al.* [66] further extend the notion of consistency from instances to clusters for multi-view graph data. More recently, Chen *et al.* [67] learn view-invariant representations by contrasting cluster assignments across views.

Many of the above methods rely on multi-view or augmentation-based strategies to define positive pairs. In contrast, our method derives positive and negative pairs directly from the self-representation matrix, thereby injecting subspace-structure cues into the contrastive learning process and eliminating the need for explicit data augmentation.

III. METHODOLOGY

In this section, we present a scalable framework for clustering large-scale hyperspectral images (HSIs). As illustrated in Fig. 1, the proposed method adopts a teacher–student distillation architecture that couples model-driven self-representation with efficient assignment prediction. Specifically, an unfolded-ADMM teacher computes batch-specific self-representation coefficients $\mathbf{C}^{(b)}$ (and the induced affinities) to construct reliable positive/negative relations, while a lightweight student network is trained with a contrastive

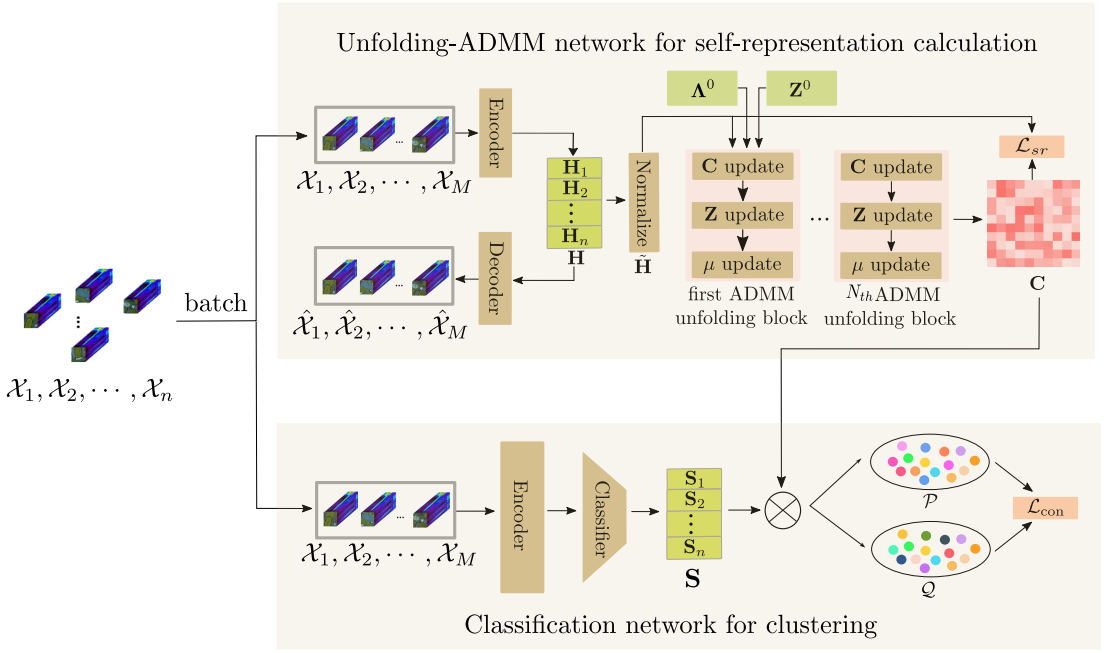


Fig. 1: Overview of the proposed framework. The unfolded-ADMM teacher is first trained and then frozen. For each mini-batch, it computes a batch-specific self-representation matrix $\mathbf{C}^{(b)}$ (and the induced affinity), from which positive and negative index sets $\mathcal{P}^{(b)}$ and $\mathcal{Q}^{(b)}$ are constructed to supervise the student. The student network is optimized to predict the soft assignments $\mathbf{S}$ via a contrastive objective. Random mini-batch sampling enables the student to progressively accumulate structural knowledge across iterations.

objective to predict the soft assignments $\mathbf{S}$ efficiently. With random mini-batch sampling, the distilled relations vary across iterations and progressively cover diverse parts of the dataset, allowing the student to accumulate global structural knowledge in its parameters.

A. Problem Formulation

Let $\mathcal{X} = \{\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_N\}$ denote the set of N overlapping patches extracted from an HSI scene $\mathcal{I}$, where each patch $\mathcal{X}_i \in \mathbb{R}^{A\times A\times B}$ is centered on a pixel and covers an $A \times A$ spatial neighborhood over all B spectral bands. Our goal is to cluster these patches into K classes in an unsupervised manner. Deep subspace clustering typically adopts a

self-representation model to capture the intrinsic structure of the latent features:

$$\min_{\Theta,\mathbf{C}} \quad \frac{1}{2N}\sum_{i=1}^{N}\left\|\mathcal{X}_i-\hat{\mathcal{X}}_i\right\|_F^2+\|\mathbf{H}-\mathbf{H}\mathbf{C}\|_F^2+\lambda\|\mathbf{C}\|_1 \tag{1}$$

$$\text{s.t.} \quad \operatorname{diag}(\mathbf{C})=\mathbf{0}, \tag{2}$$

where $\hat{\mathcal{X}}_i = D(E(\mathcal{X}_i))$ is the reconstruction produced by an autoencoder with parameters Θ. $\mathbf{H} \in \mathbb{R}^{V\times N}$ denotes the latent feature matrix (V is the feature dimension), and $\mathbf{C} \in \mathbb{R}^{N\times N}$ is the self-representation coefficient matrix whose entry c_{ij} measures the contribution of sample j in representing sample i. λ controls the sparsity of $\mathbf{C}$, and the constraint $\operatorname{diag}(\mathbf{C}) = \mathbf{0}$ prevents the trivial self-reconstruction solution.

Directly optimizing the above objective over all N samples requires constructing and updating a global coefficient matrix $\mathbf{C} \in \mathbb{R}^{N\times N}$, which incurs $\mathcal{O}(N^2)$ memory and typically super-quadratic computation, making it impractical for large-scale HSIs. To enable scalable training, we adopt a mini-batch surrogate: at each iteration, we solve for a batch-specific matrix $\mathbf{C}^{(b)} \in \mathbb{R}^{M\times M}$ using only the M samples in the current mini-batch, with $\operatorname{diag}(\mathbf{C}^{(b)}) = \mathbf{0}$. This strategy avoids constructing the global $N\times N$ matrix and yields $\mathcal{O}(M^2)$ memory cost per mini-batch, while the per-batch computation is dominated by solving an $M \times M$ linear system in the unfolded ADMM module. Over one epoch (approximately N/M mini-batches), the overall cost scales linearly with N for fixed M. The resulting batch-wise coefficients are then distilled into contrastive supervision (i.e., relational distillation): samples with large coefficients are treated as positives, while samples with small coefficients (optionally filtered by low cosine similarity) are treated as negatives. Across training, random mini-batch sampling progressively exposes the student network to diverse local relational cues, enabling it to learn a global embedding and produce efficient cluster assignments at inference.

B. Teacher Network: Self-Representation Learning

Most existing deep subspace clustering methods parameterize self-representation using a fully connected layer in latent space, treating the coefficient matrix as globally shared network weights. This design is inherently tied to full-batch training, since the self-representation matrix is defined over the specific set of samples involved in the optimization. Under mini-batch training, however, different batches correspond to different sets of samples and therefore require different coefficient matrices, making a globally shared parameterization unsuitable.

To address this issue, we adopt a batch-specific unfolded ADMM module that computes self-representation coefficients independently at each training step. An autoencoder acts as a global nonlinear feature extractor, producing latent representations for each mini-batch. Crucially, the coefficient matrix is not stored as trainable network weights; instead, it is computed on-the-fly via a fixed number of unfolded ADMM updates using only the features within the current mini-batch. As a result, each iteration yields a batch-specific coefficient matrix, avoiding coefficient reuse across batches and enabling self-representation learning in a scalable mini-batch regime.

a) ADMM-Based Differentiable Solver: To solve the batch-wise sparse self-representation problem, we adopt ADMM by introducing an auxiliary variable $\mathbf{Z}^{(b)}$ and enforcing $\mathbf{C}^{(b)} = \mathbf{Z}^{(b)}$. This yields the augmented Lagrangian

$$\begin{aligned}\mathcal{L}\Big(\mathbf{C}^{(b)}, \mathbf{Z}^{(b)}, \mathbf{\Lambda}^{(b)}\Big) = \frac{1}{2}\left\|\tilde{\mathbf{H}}^{(b)} - \tilde{\mathbf{H}}^{(b)}\mathbf{C}^{(b)}\right\|_F^2 + \lambda\left\|\mathbf{Z}^{(b)}\right\|_1 \\ + \left\langle \mathbf{\Lambda}^{(b)},\, \mathbf{C}^{(b)} - \mathbf{Z}^{(b)} \right\rangle + \frac{\rho}{2}\left\|\mathbf{C}^{(b)} - \mathbf{Z}^{(b)}\right\|_F^2,\end{aligned} \tag{3}$$

where $\mathbf{\Lambda}^{(b)}$ is the Lagrange multiplier and $\rho > 0$ is the penalty parameter. The variables are updated iteratively as follows.

C-update. Given $\mathbf{Z}^{(b,t)}$ and $\mathbf{\Lambda}^{(b,t)}$, $\mathbf{C}^{(b,t+1)}$ is obtained by solving a quadratic subprob-

lem with the closed-form solution

$$\mathbf{C}^{(b,t+1)} = \left(\tilde{\mathbf{H}}^{(b)\top}\tilde{\mathbf{H}}^{(b)} + \rho\mathbf{I}_M\right)^{-1}\left(\tilde{\mathbf{H}}^{(b)\top}\tilde{\mathbf{H}}^{(b)} - \mathbf{\Lambda}^{(b,t)} + \rho\mathbf{Z}^{(b,t)}\right). \tag{4}$$

Z-update. $\mathbf{Z}^{(b,t+1)}$ is obtained by element-wise soft-thresholding:

$$\tilde{\mathbf{Z}}^{(b,t+1)} = T_{\lambda/\rho}\left(\mathbf{C}^{(b,t+1)} + \frac{1}{\rho}\mathbf{\Lambda}^{(b,t)}\right), \tag{5}$$

where $T_\kappa(\cdot)$ is applied element-wise as

$$T_\kappa(x) = \operatorname{sgn}(x)\cdot\max\left(|x| - \kappa,\, 0\right). \tag{6}$$

To prevent the trivial self-reconstruction, we enforce the zero-diagonal constraint by

$$\mathbf{Z}^{(b,t+1)} = \tilde{\mathbf{Z}}^{(b,t+1)} - \operatorname{diag}\left(\tilde{\mathbf{Z}}^{(b,t+1)}\right), \tag{7}$$

which, together with the constraint $\mathbf{C}^{(b)} = \mathbf{Z}^{(b)}$, implies $\operatorname{diag}(\mathbf{C}^{(b)}) = \mathbf{0}$ at convergence.

$\mathbf{\Lambda}$-update. The multiplier is updated using the primal residual:

$$\mathbf{\Lambda}^{(b,t+1)} = \mathbf{\Lambda}^{(b,t)} + \rho\left(\mathbf{C}^{(b,t+1)} - \mathbf{Z}^{(b,t+1)}\right). \tag{8}$$

To obtain a more accurate self-representation matrix, we unfold T iterations of the ADMM solver into a finite-depth network, enabling end-to-end learning of the feature extractor together with a small set of solver hyperparameters. In particular, the nonlinear encoder maps HSI patches into a representation space where the linear self-representation assumption is more appropriate, thereby improving the ability to capture complex structures in practice. The resulting unfolded ADMM architecture is summarized in Algorithm 1.

As shown in Algorithm 1, $\mathbf{W}^b$ and $(\mathbf{b})^b$ are computed from $\tilde{\mathbf{H}}^{(b)}$ independently for each mini-batch, whereas the penalty parameter ρ is shared across batches (and can be treated as a learnable parameter). Moreover, the soft-thresholding step is implemented using

Algorithm 1 Unfolded ADMM for Mini-batch Self-Representation (with $\frac{1}{2}$ scaling)

Require: Latent features $\tilde{\mathbf{H}}^{(b)} \in \mathbb{R}^{V \times M}$; sparsity parameter λ; number of unfolding steps T; penalty parameter ρ
Ensure: Self-representation matrix $\mathbf{C}^{(b,T)}$; auxiliary variable $\mathbf{Z}^{(b,T)}$
Initialize: $\mathbf{Z}^{(b,0)} \leftarrow 10^{-3}\mathbf{1}_{M\times M}$, $\mathbf{C}^{(b,0)} \leftarrow \mathbf{0}_{M\times M}$, $\mathbf{\Lambda}^{(b,0)} \leftarrow \mathbf{0}_{M\times M}$
Precompute (per mini-batch):
$(\mathbf{b})^b \leftarrow \left(\tilde{\mathbf{H}}^{(b)\top}\tilde{\mathbf{H}}^{(b)} + \rho\mathbf{I}_M\right)^{-1}$
$\mathbf{W}^b \leftarrow (\mathbf{b})^b\tilde{\mathbf{H}}^{(b)\top}$
for $t = 1$ to T **do**
  $\mathbf{C}^{(b,t)} \leftarrow \mathbf{W}^b\tilde{\mathbf{H}}^{(b)} - (\mathbf{b})^b\left(\mathbf{\Lambda}^{(b,t-1)} - \rho\mathbf{Z}^{(b,t-1)}\right)$
  $\tilde{\mathbf{Z}}^{(b,t)} \leftarrow \mathbf{C}^{(b,t)} + \frac{1}{\rho}\mathbf{\Lambda}^{(b,t-1)}$
  $\tilde{\mathbf{Z}}^{(b,t)} \leftarrow \mathrm{ReLU}\left(|\tilde{\mathbf{Z}}^{(b,t)}| - \lambda/\rho\right) \odot \mathrm{sgn}\left(\tilde{\mathbf{Z}}^{(b,t)}\right)$
  $\mathbf{Z}^{(b,t)} \leftarrow \tilde{\mathbf{Z}}^{(b,t)} - \mathrm{diag}\left(\tilde{\mathbf{Z}}^{(b,t)}\right)$
  $\mathbf{\Lambda}^{(b,t)} \leftarrow \mathbf{\Lambda}^{(b,t-1)} + \rho\left(\mathbf{C}^{(b,t)} - \mathbf{Z}^{(b,t)}\right)$
end for
  return $\mathbf{C}^{(b,T)}, \mathbf{Z}^{(b,T)}$

basic operations such as $\mathrm{ReLU}(\cdot)$, absolute value, and $\mathrm{sgn}(\cdot)$, which allows gradients to propagate through the unfolded iterations for end-to-end optimization.

b) Multi-Objective Loss Function: Our training objective combines five complementary components:

Reconstruction Loss: Ensure the latent representation contained enough information from the original data:

$$\mathcal{L}_{\text{recon}}^{(b)} = \frac{1}{2M}\|\mathcal{X}^{(b)} - \hat{\mathcal{X}}^{(b)}\|_F^2 \tag{9}$$

Representation Loss: Ensures accurate feature reconstruction within each batch:

$$\mathcal{L}_{\text{rep}}^{(b)} = \left\|\tilde{\mathbf{H}}^{(b)} - \tilde{\mathbf{H}}^{(b)}\mathbf{C}^{(b)}\right\|_F^2 \tag{10}$$

Sparsity Loss: Promotes sparsity to reveal meaningful local structures:

$$\mathcal{L}_{\text{sparse}}^{(b)} = \frac{1}{M}\sum_{j=1}^{M}\left\|\mathbf{C}_{:,j}^{(b)}\right\|_1 \tag{11}$$

Consistency Loss: Ensures stable ADMM convergence:

$$\mathcal{L}_{\text{consist}}^{(b)} = \left\|\mathbf{C}^{(b)} - \mathbf{Z}^{(b)}\right\|_F^2 \tag{12}$$

Block Structure Loss: Encourages clustering-friendly block-diagonal structure based on spectral graph theory:

$$\mathbf{A}_{ij}^{(b)} = \begin{cases} \frac{1}{2}\left(|\mathbf{C}_{ij}^{(b)}| + |\mathbf{C}_{ji}^{(b)}|\right), & i \neq j \\ 0, & i = j \end{cases} \tag{13}$$

$$\mathbf{G}^{(b)} = \mathbf{T}^{(b)} - \mathbf{A}^{(b)}, \quad \mathbf{T}_{ii}^{(b)} = \sum_j \mathbf{A}_{ij}^{(b)} \tag{14}$$

$$\mathcal{L}_{\text{block}}^{(b)} = \sum_{i=1}^{K} \varepsilon_i\left(\mathbf{G}^{(b)}\right) \tag{15}$$

The total objective per batch becomes:

$$\mathcal{L}_{\text{total}}^{(b)} = \mathcal{L}_{\text{recon}}^{(b)} + \beta\mathcal{L}_{\text{rep}}^{(b)} + \beta_1\mathcal{L}_{\text{sparse}}^{(b)} + \beta_2\mathcal{L}_{\text{consist}}^{(b)} + \beta_3\mathcal{L}_{\text{block}}^{(b)} \tag{16}$$

This combined loss function integrates reconstruction quality, representation accuracy, sparsity regularization, consistency constraints, and block-structure enforcement, with positive hyperparameters β, β_1, β_2, and β_3 balancing their relative importance.

C. Adaptive Sample Selection Strategy

After the teacher network has been trained, its weights are frozen. It then acts as a fixed feature extractor and a self-representation matrix generator, providing the batch-specific $\mathbf{C}^{(b)}$ to facilitate the training of the student network. To enable adaptive and effective contrastive learning for the student network, we construct high-quality positive and negative index sets using a dual-criteria strategy that leverages both the self-representation matrix $\mathbf{C}^{(b)}$ (derived from the teacher network's output for the current batch) and the cosine similarity matrix $\mathbf{R}^{(b)}$ (computed from the original features of the

current batch). Unlike methods relying on fixed sample sizes, this strategy dynamically adjusts the number of positive and negative samples based on the evolving latent structure revealed within each training batch, ensuring the samples' selection truly reflects the current data representations.

a) Dual-Criteria Sample Construction: We adopt separate constructions for positive and negative samples to balance coverage and purity.

Positive Samples (Self-Representation Cumulative Thresholding). Positive samples are selected from the self-representation affinity matrix $\mathbf{A}^{(b)}$ using an anchor-wise cumulative threshold rule. For the i-th sample in the b-th batch, we define the nonnegative affinity score vector

$$\mathbf{a}_i \triangleq |\mathbf{A}^{(b)}_{i,:}|, \quad a_{ii} = 0, \tag{17}$$

and construct the positive index set $\mathcal{P}_i^{(b)}$ as follows:

1) sort the entries of $\mathbf{a}_i$ in descending order,
2) compute the cumulative sum along the sorted list, and
3) select the top indices whose cumulative mass does not exceed $q_p \cdot \sum_{j=1}^{M} a_{ij}$.

We set $q_p = 0.9$ to preserve dominant contributors while suppressing weak and noisy connections.

Negative Samples (Intersection Strategy). To obtain high-confidence negatives, we intersect weak-affinity candidates and feature-space validators:

$$\mathcal{Q}_i^{(b)} = \mathcal{C}_i^{(b)} \cap \mathcal{F}_i^{(b)}. \tag{18}$$

Here, $\mathcal{C}_i^{(b)}$ is derived from $\mathbf{a}_i$ by selecting weak connections: we select the indices in the tail of the sorted list such that their cumulative mass is at most q_n of the total (i.e., the weakest connections). Meanwhile, the feature-space validator is defined as the index set

$$\mathcal{F}_i^{(b)} = \{\, j \mid j \neq i,\ \mathbf{R}_{ij}^{(b)} < \delta \,\}, \tag{19}$$

where $\mathbf{R}^{(b)}$ denotes the cosine similarity matrix computed on the current batch and we set $\delta = 0.2$. We use $q_n = 0.05$ to introduce a margin and reduce overlap with ambiguous samples.

D. Student Network: Contrastive Clustering Objective

The student network comprises an encoder followed by a K-dimensional clustering assignment head, which produces soft cluster assignments $\mathbf{S}^{(b)} \in \mathbb{R}^{M\times K}$ through a softmax activation. To transfer structural knowledge from the teacher network, the student is trained using positive and negative index sets $(\mathcal{P}_i^{(b)}, \mathcal{Q}_i^{(b)})$ constructed from the teacher-guided relations for the current batch.

We compute the similarity between two samples i and j using the exponentiated cosine similarity of their assignment vectors:

$$\hat{\mathbf{s}}_i^{(b)} = \frac{\mathbf{s}_i^{(b)}}{\|\mathbf{s}_i^{(b)}\|_2}, \quad G(i,j) = \exp\left(\frac{(\hat{\mathbf{s}}_i^{(b)})^\top \hat{\mathbf{s}}_j^{(b)}}{\tau}\right), \tag{20}$$

where $\mathbf{s}_i^{(b)}$ and $\mathbf{s}_j^{(b)}$ denote the i-th and j-th rows of $\mathbf{S}^{(b)}$, respectively, and τ is the temperature parameter.

We adopt a contrastive loss based on the InfoNCE formulation to encourage samples from the same cluster to exhibit high similarity, while discouraging similarity between those from different clusters:

$$\mathcal{L}_{\text{con}} = \frac{1}{M}\sum_{i=1}^{M} -\log\left(\frac{\sum\limits_{j\in\mathcal{P}_i^{(b)}} G(i,j)}{\sum\limits_{j\in\mathcal{P}_i^{(b)}} G(i,j) + \sum\limits_{j\in\mathcal{Q}_i^{(b)}} G(i,j) + \epsilon}\right) \tag{21}$$

where ϵ is a small constant added to avoid numerical instability when the denominator approaches zero. This contrastive objective alone provides effective supervision, enabling the student model to recover the clustering structure distilled from the teacher network.

E. Training Strategy

The training process consists of two stages. In the first stage, the unfolded-ADMM teacher is trained using the multi-objective loss in Sec. III-B to produce stable batch-wise self-representation signals. After training, the teacher is frozen for the remainder of the procedure. In the second stage, for each mini-batch, the teacher and student networks process the data in parallel. The teacher computes a batch-specific self-representation matrix $\mathbf{C}^{(b)}$ (and the induced affinity), which, together with the within-batch cosine similarity matrix $\mathbf{R}^{(b)}$, is used to construct the positive and negative index sets $\{\mathcal{P}_i^{(b)}\}$ and $\{\mathcal{Q}_i^{(b)}\}$. These sets provide relational supervision for the student, which is optimized using the contrastive objective in Sec. III-D.

IV. Experiments

In this section, we present a comprehensive evaluation of our method, conducting experimental analyses to demonstrate its effectiveness and robustness across various scenarios.

A. Datasets

We conduct the experiments on 3 well-known real-world HSI datasets: Houston, Urban, and Trento.

a) Houston dataset: The Houston dataset contains 144 spectral bands. A representative area with a size of 130 $\times$ 130 pixels is selected, containing 7 classes and 6,104 labeled pixels.

b) Urban dataset: The HYDICE Urban dataset contains 162 spectral bands. A representative area with a size of 150 $\times$ 160 is selected, containing 7 classes and 12,048 labeled pixels.

c) Trento dataset: The Trento dataset contains 63 spectral bands. In this work, we use the full scene with a size of 600 $\times$ 166, containing 6 classes and 30,214 labeled samples. The cluster distribution is shown in Table I.

Houston		**Urban**		**Trento**	
Class Index	Counts	Cluster Index	Counts	Cluster Index	Counts
1	1342	1	2072	1	4034
2	863	2	2637	2	2903
3	558	3	1724	3	479
4	2003	4	966	4	9123
5	130	5	1903	5	10501
6	413	6	1844	6	3174
7	795	7	902	-	-

TABLE I: Ground-truth class distribution of the labeled pixels in each dataset.

B. Parameter setting

The proposed method operates within a knowledge distillation framework, where the unfolding ADMM network serves as the Teacher, and the student network acts as the Student. The training process involves two stages: first optimizing the Teacher, and subsequently training the Student under the supervision of the Teacher. For consistency, most hyperparameters are fixed across datasets. Specifically, the learning rate is set to 1×10^{-4} for the Teacher network and 1×10^{-5} for the Student network. Regarding the loss function terms, the initial penalty parameter ρ is set to 2.0, while the balance weights are configured as follows: self-representation $\beta = 10$, L1 sparsity $\beta_1 = 1.0$, convergence constraint $\beta_2 = 1.0$, and the connectivity constraint $\beta_3 = 5$. The only dataset-dependent parameter is the sparsity weight λ, which is tuned to 0.01, 0.06, and 0.08 for the Houston, Trento, and Urban datasets, respectively.

C. Comparison Methods

We compare the proposed method with representative baselines from different clustering families, including: (i) centroid-based methods, FCM [18] and DEKM [42]; (ii) graph-based methods, spectral clustering (SC) [26] and SpectralNet [68]; (iii) hierarchical clustering, FINCH [27]; (iv) traditional sketch-based subspace clustering, Sketch-SSC-

TV [53]; (v) data-driven deep clustering, IDEC [69]; and (vi) subspace-clustering-based methods, HyperAE [45], EDESC [51], and SCDSC [70]. To ensure a fair comparison, all methods are evaluated on the same input data and under the same experimental protocol. For stochastic methods, we perform 10 runs with different random seeds on each dataset and report the mean (and standard deviation) of the results. For implementation, the code for IDEC and DEKM is obtained from the ClustPy library [71], while the remaining baselines are reproduced using their official implementations.

D. Evaluate Metrics

In our experiments, we use 3 widely used metrics to evaluate and compare the performance of different methods. Accuracy, Normalized mutual information, and Kappa, what's more, we also include the running time.

- **Overall Accuracy (OA)**: Accuracy measures the proportion of correctly classified samples over the total number of samples, providing a basic but effective evaluation of classification performance.

$$\text{OA} = \frac{\sum_{i=1}^{n} \delta(y_i, \hat{y}_i)}{n}$$

  Here, y_i is the true label of the i-th sample, $\hat{y}_i$ is the predicted label of the i-th sample, $\delta(y_i, \hat{y}_i)$ is an indicator function that equals 1 if $y_i = \hat{y}_i$ and 0 otherwise, and n is the total number of samples.

- **Normalized Mutual Information (NMI)**: NMI quantifies the similarity between the predicted clustering and the ground truth labels by measuring how much information one set of labels shares with the other, normalized to make the values comparable across datasets.

$$\text{NMI} = \frac{2 \cdot I(\mathbf{y}; \hat{\mathbf{y}})}{H(\mathbf{y}) + H(\hat{\mathbf{y}})}$$

  Here, $I(\mathbf{y}; \hat{\mathbf{y}})$ is the mutual information between the true labels $\mathbf{y}$ and the predicted

labels $\hat{\mathbf{y}}$, while $H(\mathbf{y})$ and $H(\hat{\mathbf{y}})$ are the entropies of the true and predicted labels, respectively.

The mutual information $I(\mathbf{y}; \hat{\mathbf{y}})$ is further defined as:

$$I(\mathbf{y}; \hat{\mathbf{y}}) = \sum_{y \in \mathbf{y}} \sum_{\hat{y} \in \hat{\mathbf{y}}} p(y, \hat{y}) \log \frac{p(y, \hat{y})}{p(y)p(\hat{y})}$$

Here, $p(y, \hat{y})$ is the joint probability distribution of y and $\hat{y}$, and $p(y)$, $p(\hat{y})$ are their marginal probabilities.

- **Cohen's Kappa (Kappa)**: Cohen's Kappa measures the agreement between predicted and ground truth labels while accounting for agreement occurring by chance, making it more robust than simple accuracy for imbalanced datasets.

$$\mathcal{K} = \frac{P_o - P_e}{1 - P_e}$$

Here, P_o is the observed agreement, calculated as the ratio of correctly classified samples to the total number of samples, and P_e is the expected agreement, which is computed based on the marginal probabilities of the predicted and true labels.

E. Performance Analyses

We report the clustering performance of the proposed method and competing baselines on three benchmark datasets. For stochastic methods, we repeat experiments with multiple random seeds and report the mean and the standard deviation (Std), where a smaller Std indicates better stability (i.e., less sensitivity to initialization and training randomness). For OA and $\mathcal{K}$, predicted cluster indices are aligned with ground-truth labels using the Hungarian algorithm.

a) Houston dataset: The clustering results are shown in Table II and Fig. 2.

Our method achieves the best mean OA (85.44%) and $\mathcal{K}$ (0.8163). It attains the second-best NMI (0.8501), which is only 0.0033 lower than the best result (0.8534), indicating highly comparable cluster consistency.

TABLE II: Quantitative evaluation of different clustering methods on the Houston dataset.

Class	FCM	SC	FINCH	Sketch-SSC-TV	IDEC	DEKM	SN	HyperAE	EDESC	SCDSC	Ours
1	53.50	46.50	53.50	65.61	46.50	53.50	46.52	47.89	73.77	**93.07**	90.70
2	**100**	100.00	100	40.00	100	100	89.28	71.16	100.00	99.98	100
3	60.39	0	0	54.10	0	11.63	14.46	**64.92**	7.58	19.86	47.35
4	60.41	99.70	93.36	99.58	97.04	**100**	99.19	95.77	61.08	93.43	99.58
5	0	0	0	**100.00**	0.00	44.31	19.92	78.08	17.85	20.00	74.38
6	17.19	79.90	79.42	81.55	84.94	78.21	54.96	31.27	**94.94**	76.54	41.48
7	94.09	**97.61**	83.14	74.92	95.85	29.58	69.92	86.91	96.42	83.45	76.53
OA(%)	64.66	75.20	72.74	75.11	74.44	69.87	69.97	72.67	70.46	83.55	**85.44**
OA Std(%)	0.00	0.00	0.00	1.73	0.34	4.06	5.87	3.06	7.51	3.52	3.65
NMI	0.7359	0.8249	0.7786	0.7618	0.8090	0.7768	0.7090	0.7730	0.8016	**0.8534**	0.8501
NMI Std	0.0000	0.0000	0.0000	0.0316	0.0104	0.0167	0.0449	0.0424	0.0277	0.0258	0.0217
$\mathcal{K}$	0.5797	0.6934	0.6624	0.6863	0.6846	0.6144	0.6176	0.6753	0.6469	0.7922	**0.8163**
$\mathcal{K}$ Std	0.0000	0.0000	0.0000	0.0201	0.0042	0.0581	0.0789	0.0406	0.0859	0.0454	0.0441
Time(s)	83.94	3.09	**2.14**	8.90	75.34	77.63	21.47	214.37	50.53	62.17	99.41

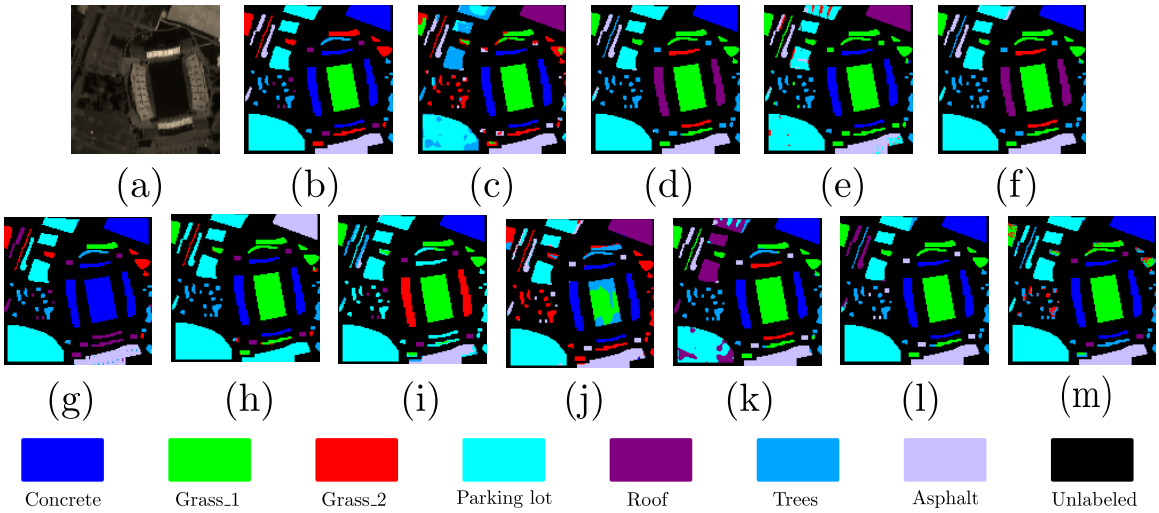


Fig. 2: Houston dataset. (a) False-color image. (b) Ground truth and the clustering results obtained by (c) FCM, (d) SC, (e) FINCH, (f) Sketch-SSC-TV, (g) IDEC, (h) DEKM, (i) SN, (j) HyperAE, (k) EDESC, (l) SCDSC, and (m) Ours.

Overall, methods that explicitly model subspace structure tend to be competitive on this dataset, outperforming several generic deep clustering baselines. The traditional Sketch-SSC-TV baseline [53] achieves a moderate OA of 75.11% on Houston, showing the efficiency of sketch-based subspace modeling but still lagging behind the strongest spatial–spectral deep subspace methods. DEKM yields relatively lower accuracy with a larger

variance (OA Std 4.06), suggesting higher sensitivity to initialization and optimization settings. SpectralNet performs substantially worse than many baselines, possibly due to training instability and the mismatch between its embedding objective and the complex spectral–spatial characteristics of HSI data.

Among subspace-based approaches, EDESC (basis representation with DEC-style refinement [40]) achieves a higher NMI than HyperAE (0.8016 vs. 0.7730), whereas HyperAE attains higher OA/$\mathcal{K}$ on this dataset. This comparison suggests that DEC-style refinement can improve mutual-information consistency, while self-representation may be more favorable for accuracy-oriented assignments in certain cases. SCDSC further improves the results, highlighting the benefit of incorporating spatial priors for HSI clustering.

In terms of efficiency, deep-learning-based methods are generally slower than non-deep baselines due to iterative training over many epochs, whereas some traditional methods (e.g., SC/FINCH) produce clustering results directly without feature learning. Sketch-SSC-TV is also efficient on Houston, requiring only 8.90 s while obtaining better OA than IDEC and FINCH. Its efficiency mainly comes from the sketching strategy: instead of constructing and optimizing a full $N \times N$ self-representation matrix over all pixels, Sketch-SSC-TV solves a reduced subspace representation problem on a much smaller sketched subset or projection. This substantially decreases the number of optimization variables and the cost of matrix operations, and it also avoids the repeated backpropagation required by deep clustering models. The TV regularization introduces spatial smoothness with limited additional overhead, so the method can remain fast while preserving some spatial consistency. However, this compressed representation may discard fine-grained spectral–spatial structures, which explains why Sketch-SSC-TV is faster but less accurate than the strongest deep spatial–spectral methods. Our method is notably more efficient than HyperAE on Houston (99.41 s vs. 214.37 s), benefiting from reduced computational burden and avoiding spectral clustering as a post-processing step. Regarding stability, smaller

TABLE III: Quantitative evaluation of different clustering methods on the Urban dataset.

Class	FCM	SC	FINCH	Sketch-SSC-TV	IDEC	DEKM	SN	HyperAE	EDESC	SCDSC	Ours
1	87.16	87.55	91.07	**93.18**	87.91	87.95	79.70	54.19	87.78	88.54	88.61
2	**100**	99.96	99.96	99.99	100.00	97.84	64.53	60.01	99.88	100	100
3	86.60	**98.72**	0	76.63	73.71	0.00	19.63	73.35	66.58	55.37	71.28
4	0.00	0	0.52	4.50	64.42	0.02	36.11	**97.01**	72.37	67.53	59.62
5	32.00	84.87	**99.84**	75.76	94.69	98.88	72.38	62.85	87.49	94.67	96.26
6	71.31	81.56	28.69	81.96	66.19	42.09	78.34	**90.99**	82.94	82.94	82.72
7	95.01	91.13	0	58.71	88.24	0.00	79.63	0.00	90.18	**99.93**	96.57
OA(%)	72.35	83.77	57.74	78.14	84.41	58.60	62.92	64.59	85.58	85.35	**87.20**
OA Std(%)	0.00	0.00	0.00	4.60	5.85	3.05	4.93	2.49	3.21	4.76	3.24
NMI	0.6819	0.7838	0.6162	0.7738	0.7890	0.6132	0.6737	0.5981	0.7851	0.8014	**0.8059**
NMI Std	0.0000	0.0000	0.0000	0.0226	0.0341	0.0164	0.0279	0.0228	0.0211	0.0234	0.0173
$\mathcal{K}$	0.6694	0.8063	0.4871	0.7385	0.8144	0.4996	0.5631	0.5838	0.8291	0.8260	**0.8478**
$\mathcal{K}$ Std	0.0000	0.0000	0.0000	0.0564	0.0699	0.0369	0.0575	0.0284	0.0387	0.0567	0.0390
Time(s)	579.88	10.85	**7.38**	15.90	161.32	159.75	45.48	650.75	111.67	104.98	212.32

Std values indicate stronger robustness to randomness, while several deep models exhibit non-negligible variance due to initialization and optimization noise.

The clustering map in Fig. 2 further shows that our predictions align most closely with the ground truth. For instance, our method reduces the confusion between the *Concrete* and *Roof* regions, which are frequently mixed by several competing methods. In addition, SCDSC produces smoother spatial patterns than EDESC, reflecting improved spatial consistency.

b) Urban dataset: The clustering results on the Urban dataset are presented in Table III and Fig. 3.

Our method achieves the best performance in terms of OA, NMI, and $\mathcal{K}$. IDEC, EDESC, and SCDSC also obtain strong results (OA above 80%). Notably, these methods employ a DEC-style refinement strategy [40], suggesting that iterative assignment refinement is particularly effective for this dataset.

Spectral clustering also performs competitively, likely benefiting from the strong local structure of the data; however, it fails to identify class 4. Sketch-SSC-TV obtains an OA of 78.14% and an NMI of 0.7738 on Urban, outperforming FINCH and several generic

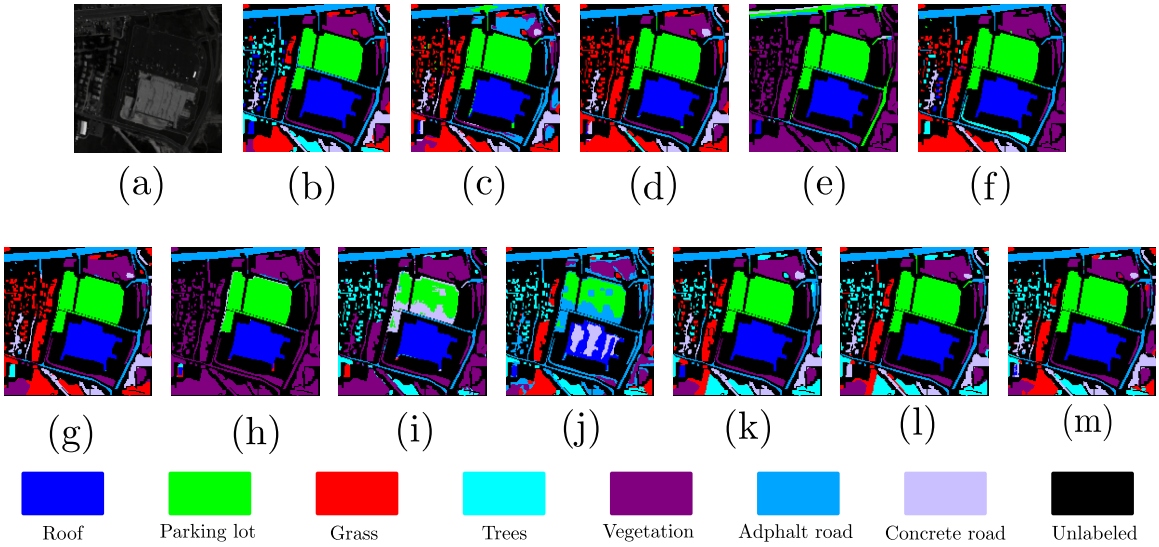


Fig. 3: Urban dataset. (a) False-color image. (b) Ground truth and the clustering results obtained by (c) FCM, (d) SC, (e) FINCH, (f) Sketch-SSC-TV, (g) IDEC, (h) DEKM, (i) SN, (j) HyperAE, (k) EDESC, (l) SCDSC, and (m) Ours.

baselines while remaining below the best DEC-style and proposed models. DEKM again shows low accuracy and non-negligible variance, indicating sensitivity to initialization and training settings. FINCH yields the lowest OA on this dataset, which may be related to unstable merges at higher hierarchy levels. SpectralNet performs worse than spectral clustering in OA, but detects all classes, indicating a more balanced partition in this setting. HyperAE achieves relatively low accuracy on Urban, which may be related to optimization difficulty and scalability on larger scenes.

In terms of runtime, HyperAE is the slowest deep baseline. Sketch-SSC-TV also keeps a low runtime on Urban (15.90 s), again benefiting from its reduced sketched optimization scale. Although the Urban dataset is roughly twice the size of Houston, HyperAE's runtime increases by about three times (650.75 s vs. 214.37 s), whereas our runtime approximately doubles (212.32 s vs. 99.41 s), indicating a more favorable empirical scaling trend.

From the clustering maps, our method, EDESC, SCDSC, IDEC, and spectral clustering produce substantially better spatial layouts than the remaining methods. For example,

TABLE IV: Quantitative evaluation of different clustering methods on the Trento dataset.

Class	FCM	SC	FINCH	Sketch-SSC-TV	IDEC	DEKM	SN	HyperAE	EDESC	SCDSC	Ours
1	66.59	0.00	1.56	91.61	99.82	0.02	75.59	–	94.29	**100.00**	99.42
2	40.82	**99.90**	81.02	28.22	58.66	0.33	62.61	–	74.28	66.91	70.45
3	15.05	82.88	0.00	**86.66**	0.21	1.63	55.91	–	20.52	10.00	38.31
4	79.67	99.96	99.97	99.93	95.82	80.00	99.93	–	99.99	**100.00**	100
5	80.98	**100.00**	99.99	76.43	95.04	100.00	87.59	–	65.97	100.00	96.97
6	80.14	24.95	80.21	70.32	73.94	**97.83**	36.39	–	71.01	73.95	78.69
OA(%)	73.67	78.47	81.36	81.43	88.70	69.25	81.43	–	80.63	92.66	**92.81**
OA Std(%)	3.09	0.29	0.00	3.13	5.81	12.10	6.68	–	4.29	1.48	2.96
NMI	0.6959	0.7544	0.7410	0.8074	0.8841	0.7406	0.7721	–	0.8429	**0.9052**	0.9027
NMI Std	0.0125	0.0061	0.0000	0.0358	0.0235	0.1244	0.0740	–	0.0294	0.0081	0.0203
$\mathcal{K}$	0.6628	0.7029	0.7425	0.7547	0.8525	0.5669	0.7481	–	0.7553	0.9020	**0.9049**
$\mathcal{K}$ Std	0.0394	0.0041	0.0000	0.0411	0.0701	0.1753	0.0918	–	0.0501	0.0197	0.0374
Time(s)	510.65	130.52	**21.89**	160.73	328.23	327.90	99.61	OOM	243.08	243.36	490.30

HyperAE tends to mix large areas of *Roof* and *Parking lot*, while DEKM and FINCH misclassify many regions as *Vegetation*. In contrast, our method more reliably separates these areas, demonstrating improved robustness to diverse spectral patterns.

c) Trento dataset: The clustering results are shown in Table IV and Fig. 4.

Our method achieves the best OA (92.81%) and $\mathcal{K}$ (0.9049), while obtaining the second-best NMI (0.9027). SCDSC achieves performance very close to ours, further confirming the importance of spectral–spatial structure in HSI clustering. IDEC and EDESC also perform well, benefiting from the DEC-style refinement strategy [40]. With the updated results, Sketch-SSC-TV reaches 81.43% OA, 0.8074 NMI, and 0.7547 $\mathcal{K}$ on Trento, indicating clear improvement over its previous run but still trailing IDEC, SCDSC, and our method in overall clustering accuracy. HyperAE is not reported on Trento due to out-of-memory (OOM) during training. Among the remaining baselines, DEKM shows the weakest overall performance with a large variance (OA Std 12.10), indicating instability under random initialization and optimization noise. From the clustering maps, our method and SCDSC resemble the ground truth most closely. IDEC and EDESC identify the main regions, while FCM captures only part of these areas and still mixes *Ground*, *Road*, and *Buildings*. Spectral clustering produces noticeable errors in the *Apple*

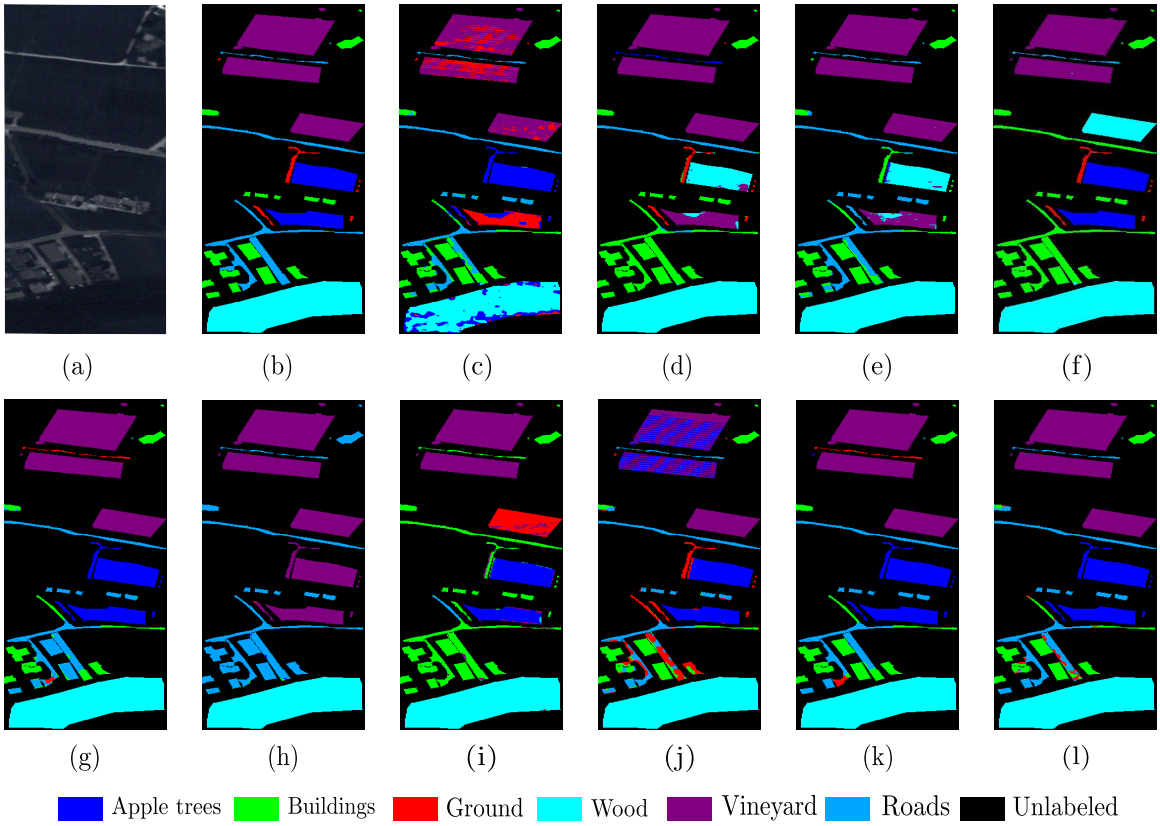
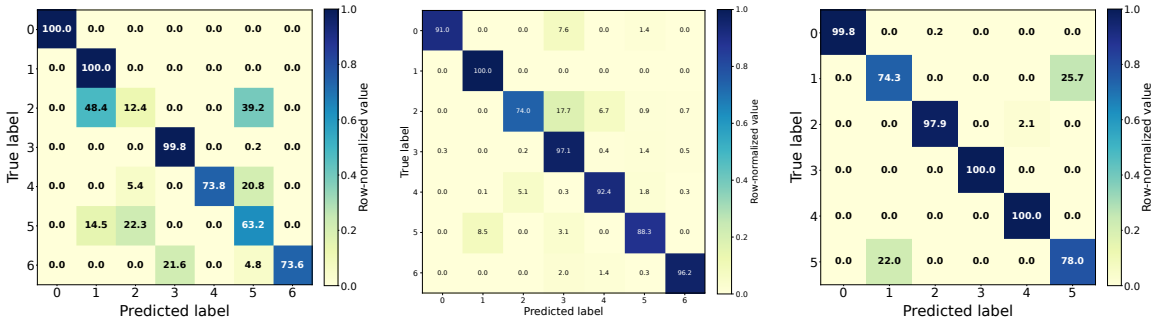


Fig. 4: Trento dataset. (a) False-color image. (b) Ground truth and the clustering results obtained by (c) FCM, (d) SC, (e) FINCH, (f) Sketch-SSC-TV, (g) IDEC, (h) DEKM, (i) SN, (j) EDESC, (k) SCDSC, and (l) Ours.

trees region. FINCH can group *Vineyard* pixels but tends to merge them with *Apple trees* at higher hierarchy levels, which degrades overall performance.

In terms of runtime, FCM (510.65 s) is the slowest on Trento, likely because iterative updates require many iterations to meet the stopping criterion. Sketch-SSC-TV runs in 160.73 s, faster than most deep baselines because sketching reduces the self-representation scale, but slower than FINCH and SpectralNet. Our method achieves the best clustering accuracy with a comparable runtime (490.30 s), suggesting that the computational cost of improved accuracy becomes more evident on larger scenes.



(a) Houston (b) Urban (c) Trento

Fig. 5: Row-normalized confusion matrices (after Hungarian matching) on three datasets.

F. Visualization of Confusion Matrix

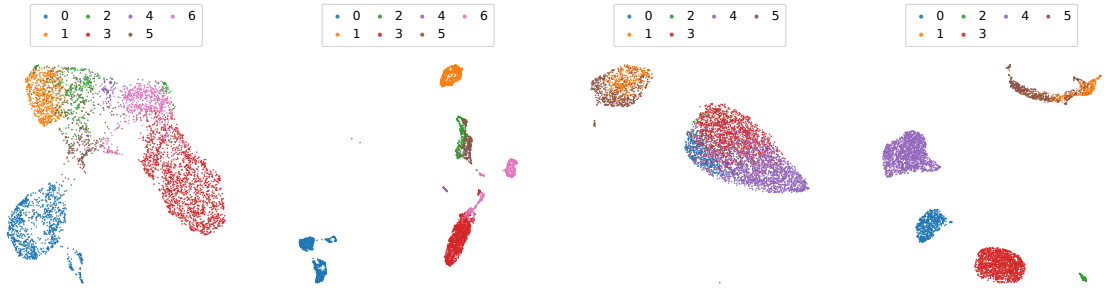
We visualize the row-normalized confusion matrices on all datasets in Fig. 5. Following Hungarian matching, each row is normalized by the number of samples in the corresponding ground-truth class, so that entries indicate per-class assignment proportions and diagonal values reflect per-class agreement after alignment. Overall, the diagonal-dominant patterns indicate that most categories are correctly separated, with the remaining errors concentrated in a few confusing class pairs.

G. Visualization of Student Output Distributions

Fig. 6 compares the student output distributions before and after optimization on the Houston and Trento datasets. The initially entangled outputs become substantially more separable after training, with tighter within-class groupings and larger inter-class margins.

H. Parameter Study

The sparsity coefficient λ controls the shrinkage strength in the self-representation module, and its optimal value can vary across datasets. As shown in Fig. 7 (evaluated on a grid $\lambda \in [0.01, 0.10]$ with step 0.01), Houston achieves the best OA at $\lambda = 0.01$ and gradually degrades as λ increases, whereas Urban benefits from stronger sparsity and



(a) Houston (before) (b) Houston (after) (c) Trento (before) (d) Trento (after)

Fig. 6: Visualization of student output distributions before and after optimization on Houston and Trento.

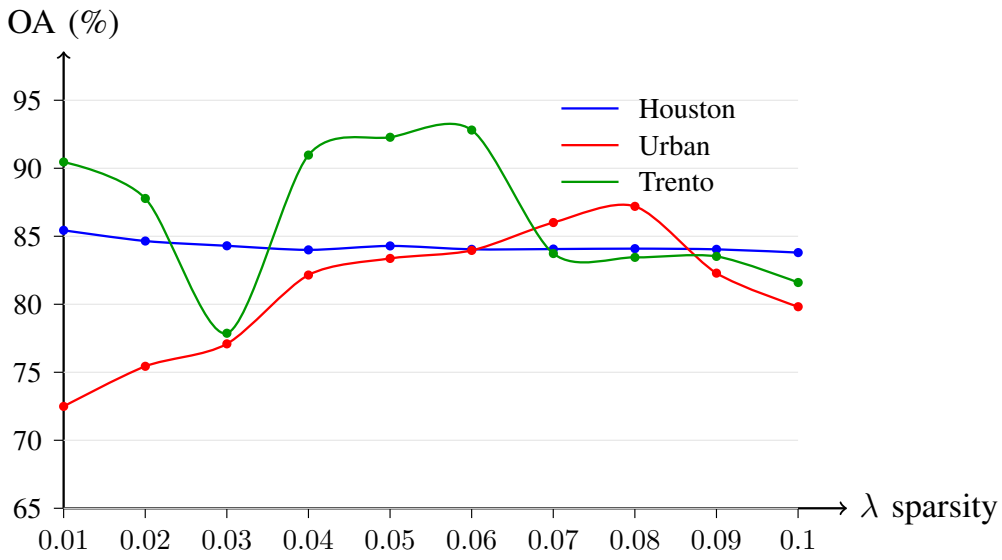
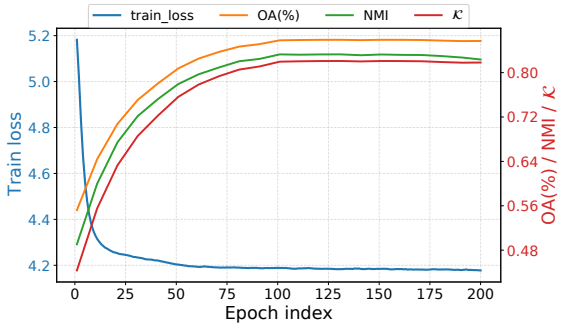


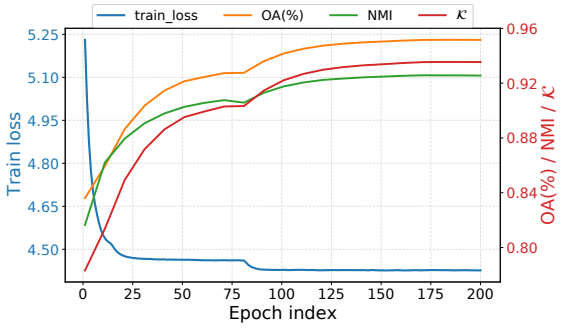
Fig. 7: Parameter study on sparsity λ. The horizontal axis ranges from 0.01 to 0.10 with a step of 0.01, and the vertical axis (ACC, i.e., OA) ranges from 65 to 95.

peaks at $\lambda = 0.08$. Trento exhibits its best performance at a moderate value ($\lambda = 0.06$), indicating a dataset-dependent trade-off.

This sensitivity is likely related to scene characteristics (e.g., class imbalance and spectral separability): overly aggressive sparsity may remove useful within-class connections and fragment minority structures, while moderate sparsity can suppress spurious cross-class links in more complex scenes. Overall, the optimal λ values across the tested datasets remain below 0.1, and typically fall within $[0.01, 0.08]$ under the tested grid.



(a) Houston dataset



(b) Trento dataset

Fig. 8: Convergence curves of two datasets.

I. Convergence Analyses

We visualize the training dynamics on two representative datasets in Fig. 8. Specifically, we plot the training objective (error metric) and the evaluation OA measured during training. As shown in Fig. 8, the train loss decreases rapidly in the early stage and then gradually plateaus, while the OA increases quickly and remains stable afterwards. These trends indicate that the proposed optimization converges efficiently and reaches a stable solution within a limited number of training epochs.

J. Ablation Study

a) The effect of the self-representation model: In our framework, positive and negative samples are constructed from the self-representation matrix produced by the proposed unfolding ADMM network. To evaluate the contribution of this learned relation modeling, we consider a heuristic baseline that builds pairs from a cosine similarity matrix using fixed thresholds: similarities above 0.7 are treated as positives and those below 0.6 as negatives. Table V reports the results. Compared with the fixed-threshold cosine strategy, self-representation-based sampling consistently improves OA, NMI, and $\mathcal{K}$ on all datasets. This suggests that the learned self-representation provides more adaptive and informative

Sample strategy	Houston			Urban			Trento		
	OA(%)	NMI	$\mathcal{K}$	OA(%)	NMI	$\mathcal{K}$	OA(%)	NMI	$\mathcal{K}$
Cosine similarity matrix	79.58	0.7857	0.7352	76.53	0.6732	0.7222	60.03	0.6515	0.5051
Self representation	**85.44**	**0.8501**	**0.8163**	**87.20**	**0.8059**	**0.8478**	**92.81**	**0.9027**	**0.9049**

TABLE V: Ablation study on sample selection strategies.

Self-representation strategy	Houston			Urban			Trento		
	OA(%)	NMI	$\mathcal{K}$	OA(%)	NMI	$\mathcal{K}$	OA(%)	NMI	$\mathcal{K}$
Classical ADMM self-representation	78.93	0.7599	0.7278	73.70	0.6384	0.6906	61.55	0.6596	0.5275
Unfolded ADMM self-representation	**85.44**	**0.8501**	**0.8163**	**87.20**	**0.8059**	**0.8478**	**92.81**	**0.9027**	**0.9049**

TABLE VI: Comparison between classical ADMM and unfolded ADMM for self-representation construction.

relational cues for sample mining, while the fixed cosine thresholds are sensitive to dataset-dependent similarity distributions and may introduce noisy positive/negative sets.

b) The Effect of Unfolded ADMM: We further evaluate the contribution of the proposed unfolded ADMM module by comparing it with a classical ADMM baseline. To ensure a fair comparison, both settings follow the same teacher–student distillation pipeline. In the classical ADMM baseline, the autoencoder is first pretrained and then fixed during the distillation stage. For each mini-batch, classical ADMM is applied to the fixed latent features to iteratively solve the self-representation problem and generate the self-representation matrix. Positive and negative samples are then constructed from this matrix and used to train the student network. Table VI reports the comparison between the classical ADMM baseline and the proposed unfolded ADMM module. The results show that the unfolded ADMM module clearly outperforms the classical ADMM baseline across all datasets and metrics. This indicates that a properly truncated unfolded ADMM module outperforms conventional ADMM solved to convergence, suggesting that joint training enables the feature representation to adapt to the finite-depth ADMM process and produce effective self-representation relations within only a few iterations.

c) The effect of learnable ρ: We further examine the role of the ADMM penalty parameter by comparing a fixed ρ with a learnable ρ that is jointly optimized during

Penalty setting	Houston			Urban			Trento		
	OA(%)	NMI	$\mathcal{K}$	OA(%)	NMI	$\mathcal{K}$	OA(%)	NMI	$\mathcal{K}$
Fixed ρ	85.21	0.8492	0.8135	86.45	0.8016	0.8390	87.52	0.8683	0.8389
Learnable ρ	**85.44**	**0.8501**	**0.8163**	**87.20**	**0.8059**	**0.8478**	**92.81**	**0.9027**	**0.9049**

TABLE VII: Effect of using a learnable penalty parameter ρ.

training. As shown in Table VII, enabling a learnable ρ improves performance across all datasets, with a particularly notable gain on Trento. This indicates that adapting the penalty strength to the data and training stage can be beneficial for stabilizing and improving the unfolded optimization process.

V. Conclusion

In this work, we presented a scalable knowledge distillation framework for hyperspectral image (HSI) subspace clustering. By adopting a teacher–student architecture, the proposed method transfers structural knowledge from an unfolded ADMM-based self-representation model to a lightweight classification network, thereby bypassing the $\mathcal{O}(N^2)$ complexity of global affinity construction. Specifically, the teacher produces mini-batch self-representations, from which an adaptive sampling strategy selects informative samples to supervise the student via contrastive learning, enabling efficient recovery of the underlying clustering structure. Experiments on three benchmark datasets demonstrate that our framework consistently improves clustering accuracy, while exhibiting near-linear runtime scaling as the data size increases. These results indicate that the proposed approach offers a practical and scalable route to deploying complex representation learning for large-scale HSI analysis.

Acknowledgment

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